

OPERATING SYSTEMS

Storage Management

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A file is a collection of related information defined by its creator

It has sequence of bits, bytes, lines or records

It is stored in the non-volatile secondary storage device such as hard disk, magnetic disk, optical disks etc

It can be used as a medium for giving input and receiving output from the program

Every file has a logical location

File Types

Executable

Object

Source Code

Batch

Text

Library

Archive

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File Concept

Unix/Linux File Types

Regular Files

Directory Files

Device Files

Link Files

Pipe Files



A File system is the data structure used to keep track of files on a disk or partition

It is responsible for organizing files and directories

It provides I/O support for a variety of storage device types

It minimizes the chances of lost

Examples: FAT16, FAT32, NTFS, Ext2, Ext3, Ext4, XFS

File Naming and Extensions

Each file is associated with a name

Name divided into 2 parts, first part the name and second part is the extension

On UNIX, extensions are not enforced by OS

However, Programming languages prefer the file extensions

Windows attaches meaning to extensions

Tries to associate applications to file extensions

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File Concept



File Type	Usual extension	Function
Executable	exe, com, bin	ready-to-run machine- language program
Object	obj, o	compiled, machine language, not linked
Source code	c. p, cpp, class	source code in various languages
Batch	bat, sh	Series of commands to be executed
Text	txt, doc	textual data documents
Word processor	doc,docs, tex, rtf	various word-processor formats
Library	lib, h	libraries of routines
Archive	arc, zip, tar	related files grouped into one file & compressed

File Attributes

Name – only information kept in human-readable form

Identifier – unique tag (number) identifies file within file system

Type – needed for systems that support different types

Location – pointer to file location on device

Size – current file size

Protection – controls who can do reading, writing, executing

Time, date, and user identification – data for protection, security, and usage monitoring

Basic File Operations

Create a file

Write to a file

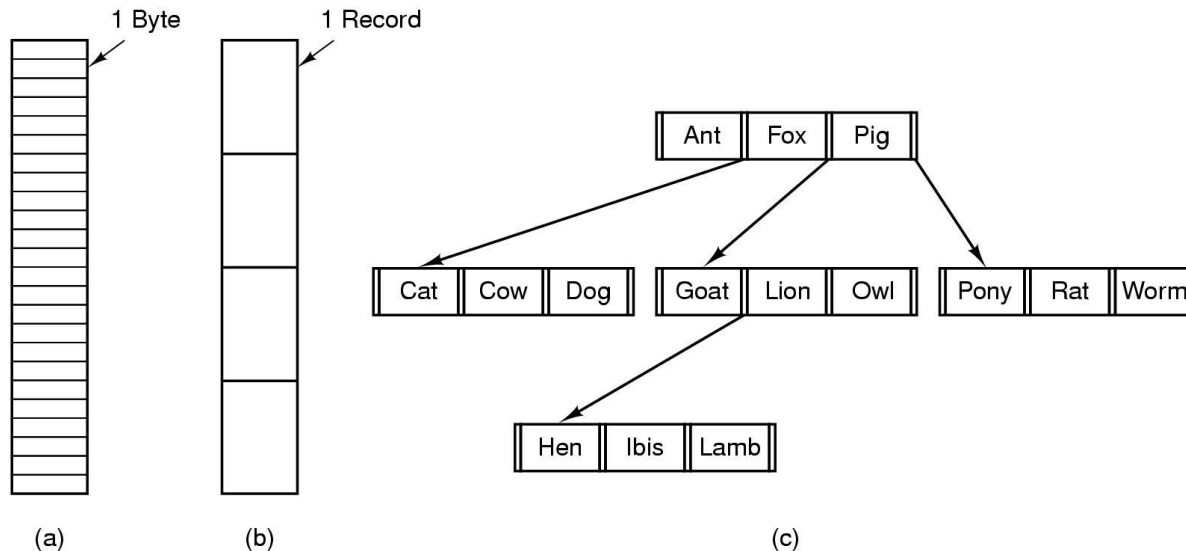
Read from a file

Seek to somewhere in a file

Delete a file

Internal File Structure

- (a) Byte Sequence: unstructured
- (b) Record sequence: r/w in records, relates to sector sizes
- (c) Complex structures, e.g. tree
 - Data stored in variable length records





THANK YOU

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